Series Solutions of Differential Equations

Prepared by Staples Education

Unit 13: Power Series, the Frobenius Method, and the Classical Special Functions

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Unit 13 at a Glance: Series Solutions

Most differential equations have no closed-form elementary solution. The series method assumes the solution is an (possibly singular) analytic function $y = \sum a_n x^n$, substitutes it into the equation, and matches coefficients of each power of x to extract a recurrence for the a_n. At an ordinary point a plain power series works; at a regular singular point the Frobenius factor x^r is required. Applied to the great equations of mathematical physics, the method manufactures the classical special functions.

- **Taylor foundations** — the coefficient formula $f^{(n)}(a)/n!$, the standard library ($e^x, \sin, \cos, \frac{1}{1-x}$), and the radius of convergence.
- **Power series solutions** — ordinary points, the ansatz $y = \sum a_n x^n$, re-indexing the derivatives, and reading off a recurrence.
- **The Frobenius method** — regular singular points, the ansatz $x^r \sum a_n x^n$, the indicial equation, and its three root cases.
- **Legendre and Bessel** — the two flagship equations, polynomial versus transcendental solutions, and orthogonality.
- **Hermite, Laguerre, Chebyshev** — three more orthogonal families, their weights, recurrences, and physical homes.

13.1 Taylor Series and the Exponential Function

Taylor and Maclaurin expansions, the coefficient formula $f^{(n)}(a)/n!$, the standard library of series, and the radius of convergence via the ratio test.

Taylor / Maclaurin Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n, \qquad \text{Maclaurin: } a = 0.$$

The Standard Library

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \ (|x| < 1).$$

Radius of Convergence (Ratio Test)

$$\frac{1}{R} = \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right|, \qquad \text{series converges for } |x-a| < R.$$

Method — Build a Maclaurin Series

1. Compute the derivatives $f^{(n)}(0)$ at the center.
2. Form the coefficients $a_n = f^{(n)}(0)/n!$.
3. Assemble the series $\sum_{n\geq 0} a_n x^n$.
4. Determine the radius of convergence with the ratio test.

Worked Example — Maclaurin Series of e^x

Build the series for $f(x) = e^x$ about $x = 0$ and read off the cubic polynomial.

$$f^{(n)}(x) = e^x \ \Rightarrow \ f^{(n)}(0) = 1 \text{ for all } n$$

$$a_n = \frac{f^{(n)}(0)}{n!} = \frac{1}{n!}$$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\frac{1}{R} = \lim_{n\to\infty} \frac{1/(n+1)!}{1/n!} = \lim_{n\to\infty} \frac{1}{n+1} = 0 \ \Rightarrow \ R = \infty$$

Key Takeaway

An analytic function equals its own Taylor series inside the radius of convergence R. This is the licence for the entire unit: if a solution is analytic, positing $y = \sum a_n x^n$ loses nothing.

Common Pitfall

A Taylor series only represents f *inside* its radius of convergence. The geometric series $\sum x^n$ equals $\frac{1}{1-x}$ only for $|x| < 1$; plugging in $x = 2$ is meaningless even though $\frac{1}{1-x}$ is perfectly finite there.

13.2 Power Series Solutions at an Ordinary Point

Ordinary points, the ansatz $y = \sum a_n x^n$, term-by-term differentiation and re-indexing, and extracting a recurrence relation for the coefficients.

The Ansatz and Its Derivatives (re-indexed to x^n)

$$y = \sum_{n=0}^{\infty} a_n x^n, \quad y' = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n, \quad y'' = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n.$$

Ordinary Point Test

$$\text{For } y'' + p(x)y' + q(x)y = 0, \quad x_0 \text{ is ordinary} \iff p, q \text{ are both analytic at } x_0.$$

Guaranteed Radius

$$R \geq \text{distance from } x_0 \text{ to the nearest singular point in the complex plane.}$$

Method — Series Solution at an Ordinary Point

1. Confirm x_0 is an ordinary point (p, q analytic there).
2. Substitute $y = \sum a_n x^n$ and the re-indexed derivatives into the equation.
3. Shift indices so every term carries the common power x^n.
4. Match coefficients of x^n to obtain a recurrence relation.
5. Solve the recurrence; a_0 and a_1 remain free and generate the two independent solutions.

Worked Example — Solve $y'' + y = 0$ by Series

Find a power series solution about the ordinary point $x = 0$ and identify the result.

$$y = \sum_{n=0}^{\infty} a_n x^n, \quad y'' = \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n$$

$$y'' + y = \sum_{n=0}^{\infty} \big[(n+2)(n+1)a_{n+2} + a_n\big]x^n = 0$$

$$a_{n+2} = \frac{-a_n}{(n+2)(n+1)}$$

$$a_0 : \ 1 - \tfrac{x^2}{2!} + \tfrac{x^4}{4!} - \cdots = \cos x; \quad a_1 : \ x - \tfrac{x^3}{3!} + \cdots = \sin x$$

$$y = a_0 \cos x + a_1 \sin x$$

Key Takeaway

At an ordinary point the two free constants a_0 and a_1 are precisely the two arbitrary constants of a second-order ODE — the even-indexed and odd-indexed coefficients build the two independent solutions.

Common Pitfall

You must *re-index* every series to the same power x^n before matching coefficients. Skipping the index shift after differentiating (forgetting the $(n+1)$ or $(n+2)(n+1)$ factor) is the single most common source of a wrong recurrence.

13.3 The Frobenius Method

Regular singular points, the Frobenius ansatz $x^r \sum a_n x^n$, the indicial equation $r(r-1)+p_0r+q_0 = 0$, and the three cases of its roots.

Regular Singular Point

x_0 is singular, yet $(x - x_0)\,p(x)$ and $(x - x_0)^2 q(x)$ are analytic at x_0.

Frobenius Ansatz and Indicial Equation

$$y = x^r \sum_{n=0}^{\infty} a_n x^n, \qquad r(r-1) + p_0\,r + q_0 = 0, \quad p_0 = \lim_{x\to 0} x\,p(x), \ q_0 = \lim_{x\to 0} x^2 q(x).$$

The Three Root Cases ($r_1 \geq r_2$)

(i) $r_1 - r_2 \notin \mathbb{Z}$: two clean Frobenius series.
(ii) $r_1 = r_2$: second solution needs a $\ln x$ term.
(iii) $r_1 - r_2 \in \mathbb{Z}^+$: $\ln x$ may appear.

Method — The Frobenius Procedure

1. Confirm x_0 is a regular singular point (xp and x^2q analytic).
2. Compute $p_0 = \lim x\,p(x)$ and $q_0 = \lim x^2 q(x)$.
3. Form and solve the indicial equation $r(r-1) + p_0 r + q_0 = 0$ for $r_1 \geq r_2$.
4. Substitute $y = \sum a_n x^{n+r}$ with the larger root to get a recurrence for a_n.
5. Build the second solution according to the root case (a log term if roots coincide or differ by an integer).

Worked Example — Indicial Roots of $x^2y'' + xy' - y = 0$

Classify $x = 0$, find p_0, q_0, and solve the indicial equation.

$$\text{Standard form: } y'' + \tfrac{1}{x}y' - \tfrac{1}{x^2}y = 0, \quad p(x) = \tfrac{1}{x},\; q(x) = -\tfrac{1}{x^2}$$

$$p_0 = \lim_{x\to 0} x \cdot \tfrac{1}{x} = 1, \qquad q_0 = \lim_{x\to 0} x^2 \cdot \left(-\tfrac{1}{x^2}\right) = -1$$

$$r(r-1) + (1)r + (-1) = r^2 - 1 = 0$$

$$r = 1 \text{ and } r = -1 \;\Rightarrow\; y = c_1 x + c_2 x^{-1}$$

Key Takeaway

The Frobenius exponent x^r is exactly what captures the singular behavior a plain power series cannot. The indicial equation comes from the *lowest* power of x in the substitution, and its roots set the leading exponents of the two solutions.

Common Pitfall

When the indicial roots are equal, or differ by a positive integer, the two series can collide. Do not assume two clean series — the smaller-root solution may require a logarithmic term $y_1 \ln x$ for independence.

13.4 Legendre and Bessel Equations

The two flagship equations of mathematical physics, the contrast between terminating polynomial solutions and transcendental ones, and orthogonality.

Legendre's Equation

$$(1-x^2)y'' - 2xy' + n(n+1)y = 0, \quad \text{singular at } x = \pm 1,\; \text{ordinary at } x = 0.$$

Legendre Polynomials and Orthogonality

$$P_0 = 1,\; P_1 = x,\; P_2 = \tfrac{1}{2}(3x^2 - 1); \qquad \int_{-1}^{1} P_m(x) P_n(x)\, dx = \frac{2}{2n+1}\,\delta_{mn}.$$

Bessel's Equation of Order ν

$$x^2y'' + xy' + (x^2 - \nu^2)y = 0, \quad \text{regular singular at } x = 0, \text{ indicial roots } r = \pm\nu.$$

Method — Recognize and Solve a Named Equation

1. Match the equation to its standard form and read off the parameter (n or ν).
2. Classify $x = 0$: Legendre is ordinary there (use a power series); Bessel is regular singular (use Frobenius).
3. For Legendre with integer n, the series terminates into the degree-n polynomial P_n.
4. For Bessel, the larger root $+\nu$ gives the bounded first-kind solution J_ν.

Worked Example — The Legendre Polynomial P_2

For $n = 2$, Legendre's equation $(1 - x^2)y'' - 2xy' + 6y = 0$ has a terminating solution. Normalize so $P_2(1) = 1$.

$$\text{Even series: } a_2 = -\tfrac{n(n+1)}{2}a_0 = -\tfrac{6}{2}a_0 = -3a_0, \quad a_4 = 0 \text{ (terminates)}$$

$$y = a_0(1 - 3x^2)$$

$$\text{Normalize } P_2(1) = 1: \; a_0(1 - 3) = -2a_0 = 1 \Rightarrow a_0 = -\tfrac{1}{2}$$

$$P_2(x) = -\tfrac{1}{2}(1 - 3x^2) = \tfrac{1}{2}(3x^2 - 1)$$

Key Takeaway

The special functions are simply the named solutions of the physically important singular ODEs. Legendre arises from spherical symmetry and terminates into polynomials for integer n; Bessel arises from cylindrical symmetry and gives the oscillatory J_ν.

Common Pitfall

Bessel's equation at $x = 0$ is a *regular singular point*, not an ordinary point — the coefficients blow up there. A plain power series fails; you must use Frobenius with indicial roots $r = \pm\nu$.

13.5 Hermite, Laguerre, and Chebyshev Polynomials

Three more classical orthogonal families: their differential equations, recurrences, weight functions, and the physical or numerical settings where each appears.

Hermite (physicists' convention)

$$y'' - 2xy' + 2ny = 0; \quad H_0 = 1, \; H_1 = 2x, \; H_2 = 4x^2 - 2; \quad \text{weight } e^{-x^2} \text{ on } (-\infty, \infty).$$

Laguerre

$$xy'' + (1-x)y' + ny = 0; \quad \text{weight } e^{-x} \text{ on } [0, \infty).$$

Chebyshev (first kind)

$$(1-x^2)y'' - xy' + n^2 y = 0; \quad T_n(\cos\theta) = \cos(n\theta)$$
$$T_0 = 1,\ T_1 = x,\ T_2 = 2x^2 - 1; \quad \text{weight } \frac{1}{\sqrt{1-x^2}}.$$

Method — Place a Classical Family

1. Inspect the leading coefficient and the first-derivative term to fingerprint the equation.
2. Match to the family: constant lead with $-2xy'$ is Hermite; xy'' with $(1-x)y'$ is Laguerre; $(1-x^2)y''$ with a single xy' is Chebyshev.
3. Generate the polynomials from the family's recurrence.
4. Pair with the correct weight function for orthogonality.

Worked Example — H_2 and T_2 from Recurrences

Use the Hermite recurrence $H_{n+1} = 2xH_n - 2nH_{n-1}$ and the Chebyshev recurrence $T_{n+1} = 2xT_n - T_{n-1}$ to build the degree-2 members.

$$H_2 = 2xH_1 - 2(1)H_0 = 2x(2x) - 2(1) = 4x^2 - 2$$

$$T_2 = 2xT_1 - T_0 = 2x(x) - 1 = 2x^2 - 1$$

$$\text{Contrast: } H_2 = 4x^2 - 2, \quad T_2 = 2x^2 - 1, \quad P_2 = \tfrac{1}{2}(3x^2 - 1)$$

Key Takeaway

Each classical family is an orthogonal basis under its own weight, tailored to a geometry: Hermite (e^{-x^2}) to the quantum harmonic oscillator, Laguerre (e^{-x}) to the hydrogen atom, Chebyshev ($1/\sqrt{1-x^2}$) to minimax approximation.

Common Pitfall

Do not confuse the degree-2 members: $H_2 = 4x^2-2$ (Hermite), $T_2 = 2x^2-1$ (Chebyshev), and $P_2 = \frac{1}{2}(3x^2-1)$ (Legendre) all look similar but come from different equations, recurrences, and weights.

Practice Problems

Work the following ten problems in order. They climb through Taylor coefficients and the standard library, the radius of convergence, a power-series solution at an ordinary point and the re-indexing that produces its recurrence, the Frobenius classification and indicial equation, and finally the classical special functions — Legendre, Bessel, and the recurrence-built Hermite and Chebyshev polynomials. Solutions follow in Part III.

1. **(Maclaurin series)** Write the Maclaurin series of e^x and state the coefficient of x^4.
2. **(Standard library)** Write the Maclaurin series of $\cos x$ in summation form.
3. **(Radius of convergence)** Use the ratio test to find the radius of convergence of $\sum_{n=0}^{\infty} \dfrac{x^n}{2^n}$.
4. **(Series at an ordinary point)** Substitute $y = \sum a_n x^n$ into $y'' + y = 0$, derive the recurrence for a_{n+2}, and identify the two independent solutions.
5. **(Re-indexing)** For $y = \sum_{n=0}^{\infty} a_n x^n$, write y'' re-indexed so every term carries the common power x^n.
6. **(Frobenius classification)** For $x^2 y'' + xy' - y = 0$, show $x = 0$ is a regular singular point, compute p_0 and q_0, and solve the indicial equation.
7. **(Indicial roots)** Given $p_0 = \frac{3}{2}$ and $q_0 = -\frac{1}{2}$, solve the indicial equation $r(r-1) + p_0 r + q_0 = 0$.
8. **(Legendre polynomial)** For $n = 2$, Legendre's equation has a terminating solution. Find it and normalize so $P_2(1) = 1$.
9. **(Bessel's equation)** State Bessel's equation of order ν, classify the point $x = 0$, and give the indicial roots.
10. **(Classical families, capstone)** Using the Hermite recurrence $H_{n+1} = 2xH_n - 2nH_{n-1}$ and the Chebyshev recurrence $T_{n+1} = 2xT_n - T_{n-1}$ (with $H_0 = 1, H_1 = 2x, T_0 = 1, T_1 = x$), compute H_2 and T_2, and contrast them with P_2.

Complete Solutions

Solution 1. Every derivative of e^x equals e^x, which is 1 at $x = 0$, so $a_n = f^{(n)}(0)/n! = 1/n!$:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \cdots$$

The coefficient of x^4 is $\dfrac{1}{4!} = \dfrac{1}{24}$.

Solution 2. Only even powers survive, with alternating sign:

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

Solution 3. With $a_n = 1/2^n$, the ratio test gives

$$\frac{1}{R} = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{1/2^{n+1}}{1/2^n} = \frac{1}{2} \implies R = 2.$$

Solution 4. Re-index $y'' = \sum_{n\geq 0}(n+2)(n+1)a_{n+2}x^n$ and add y:

$$\sum_{n=0}^{\infty} \big[(n+2)(n+1)a_{n+2} + a_n\big]x^n = 0 \implies a_{n+2} = \frac{-a_n}{(n+2)(n+1)}.$$

The a_0 branch builds $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots = \cos x$ and the a_1 branch builds $\sin x$, so

$$y = a_0 \cos x + a_1 \sin x.$$

Solution 5. Differentiate twice and shift the index by two so the power is x^n:

$$y'' = \sum_{n=0}^{\infty} (n+2)(n+1)\, a_{n+2}\, x^n.$$

Forgetting the $(n+2)(n+1)$ factor here is the most common source of a wrong recurrence.

Solution 6. In standard form $y'' + \frac{1}{x}y' - \frac{1}{x^2}y = 0$, so $p = \frac{1}{x}$, $q = -\frac{1}{x^2}$. Since $xp = 1$ and $x^2q = -1$ are analytic at 0, the point is a regular singular point. Then

$$p_0 = \lim_{x\to 0} x\,p = 1, \qquad q_0 = \lim_{x\to 0} x^2 q = -1,$$

$$r(r-1) + r - 1 = r^2 - 1 = 0 \implies r = \pm 1.$$

Solution 7. Substitute into the indicial equation and clear the fraction:

$$r(r-1) + \tfrac{3}{2}r - \tfrac{1}{2} = r^2 + \tfrac{1}{2}r - \tfrac{1}{2} = 0 \implies 2r^2 + r - 1 = 0.$$

Factor $(2r-1)(r+1) = 0$:

$$r = \tfrac{1}{2} \quad \text{and} \quad r = -1.$$

Solution 8. For $n = 2$ the even series terminates: $a_2 = -\frac{n(n+1)}{2} a_0 = -3a_0$ and $a_4 = 0$, so $y = a_0(1 - 3x^2)$. Normalize with $P_2(1) = 1$:

$$a_0(1-3) = -2a_0 = 1 \implies a_0 = -\tfrac{1}{2},$$

$$P_2(x) = -\tfrac{1}{2}(1 - 3x^2) = \tfrac{1}{2}(3x^2 - 1).$$

Solution 9. Bessel's equation of order ν is

$$x^2 y'' + xy' + (x^2 - \nu^2)y = 0.$$

At $x = 0$ the coefficients blow up but xp and $x^2 q$ are analytic, so it is a regular singular point; the indicial equation $r^2 - \nu^2 = 0$ gives

$$r = \nu \quad \text{and} \quad r = -\nu.$$

Solution 10. Apply each recurrence once:

$$H_2 = 2xH_1 - 2(1)H_0 = 2x(2x) - 2 = 4x^2 - 2,$$

$$T_2 = 2xT_1 - T_0 = 2x(x) - 1 = 2x^2 - 1.$$

All three degree-2 members look alike but come from different equations and weights:

$$H_2 = 4x^2 - 2, \quad T_2 = 2x^2 - 1, \quad P_2 = \tfrac{1}{2}(3x^2 - 1).$$

Unit Key Takeaway

Every problem above runs the same engine: assume an analytic (or Frobenius) ansatz, substitute, and match powers to extract a recurrence. The ordinary-point case gives the two free constants a_0, a_1; the regular-singular case needs the x^r factor and its indicial roots; and the named physical equations terminate or specialize into the classical orthogonal families.